

The Effect of Expected Income on Individual Migration Decisions

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Roadmap (Talk-First Design)

- ① Motivation & Stylized Facts
- ② Model Intuition
- ③ Estimation Idea
- ④ Results: What Drives Migration?
- ⑤ Interpretation & Policy Angles
- ⑥ Q&A (Appendix: all proofs/derivations/robustness)

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- 1 Motivation & Facts
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- 6 Conclusion

Question & Motivation

- **Question:** Do expected income differences drive interstate migration?
- Migration is common for the young; repeat and return moves are pervasive.
- Policy relevance: local shocks, subsidies, welfare differentials.

Models of Migration

- Migration models are used to estimate how and why agents move within an economy
- Typically used to look at isolated migration events, which may be too shallow of a viewpoint

Contribution

- **What's new:** Develops a tractable econometric model focusing on expected income's influence on migration which allows for dynamic decision making
- **Why it matters:** Adds possibility of both repeat and return migration, creating a context for this model to solve a job-search problem
- Using data from NLSY, gain an understanding of the motives behind and patterns exhibited by movers in the framework of this model

Paper's Story / Key Findings

- The paper aims to look at white men with high school education, utilizing the NLSY to track agents through time
- Expected income prospects are a major contributor to interstate migration decisions
- Further, there is evidence of geographic factors taking place once income realization is met, and the current location is found to be suboptimal
- **Key quantitative takeaways**
 - Value of moving from a bad location to a good one equates to a \$17,000 per year
 - Moving is mostly decided by fit of draw: if good, they stay; if bad: they try again, incurring a cost

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Model Assumptions

- Assume wages are a local product of skill bundles
- J locations
- y_{ij} is individual i 's income in location j with a known distribution
- Migration occurs to maximize expected discounted value of lifetime utility
- Ignore assets for this case, only seeking to maximize expected lifetime income, net of moving costs

Value Function

$$V(x, \zeta) = \max_j (v(x, j) + \zeta_j) \quad (1)$$

where,

$$v(x, j) = u(x, j) + \beta \sum_{x'} p(x'|x, j) \bar{v}(x')$$

and

$$\bar{v}(x) = \mathbb{E}_{\zeta} V(x, \zeta)$$

Value Function, cont.

Further,

$$\exp(\bar{v}(x)) = \exp(\bar{\gamma}) \sum_{j=1}^J \exp(v(x, j)),$$

and

$$\rho(x, j) = \exp(\bar{\gamma} + v(x, j) - \bar{v}(x))$$

where $\bar{\gamma}$ is the Euler constant, $\rho(x, j)$ is the probability of choosing j when in state x

Wages

Wage of agent i in location j at age a during time t is as follows,

$$w_{ij}(a) = \mu_j + v_{ij} + G(X_i, a, t) + \eta_i + \epsilon_{ij}(a) \quad (2)$$

- Quality of match governed by the difference between $\mu_j + v_{ij}$ and $\mu_k + v_{ik}$, the current quality versus the prospective quality of another location
- Match quality is known for current and, where possible, most recent location
- Prospects of all other locations are unknown

State Variables & Flow Payoffs

- $l = (l^0, l^1, \dots, l^{M-1})$ be an M vector containing the sequence of previous locations, beginning with the current location
- ω is an M vector containing wage and utility information at these locations
- $l, \omega, a \in x$, the state vector

$$\tilde{u}_h(x, j) = u_h(x, j) + \zeta_j, \quad (3)$$

where,

$$\begin{aligned} u_h(x, j) = & \alpha_0 w(l^0, \omega) + \sum_{k=1}^K \alpha_k Y_k(l^0) + \alpha^H (l^0 = h) \\ & + \xi(l^0, \omega) - \Delta_\tau(x, j) \end{aligned}$$

Moving Costs

- $D(l^0, j)$ represents distance from current location to some other location j
- $\mathbb{A}(l^0)$ set of locations adjacent to l^0

Thus, define moving cost as

$$\begin{aligned}\Delta_\tau(x, j) = & (\gamma_{0,\tau} + \gamma_1 D(l^0, j) - \gamma_2(j \in \mathbb{A}(l^0)) \\ & - \gamma_3(j = l^1) + \gamma_4 a - \gamma_5 n_j)(j \neq l^0)\end{aligned}\tag{4}$$

Value of Moving

State at t :

current location ℓ_0 , known w_{ℓ_0}

prob. $v(\text{stay})$

prob. $\mathbb{E}(v(j))$

Stay

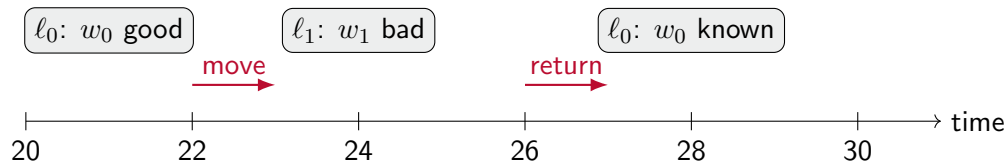
Move to $j \neq \ell_0$

$$u_t = \alpha_0 w_{\ell_0} + \text{amenities} + \zeta_{\ell_0} \\ + \beta[V_{t+1}|\text{age} + 1, \ell_0]$$

$$u_t = \alpha_0 w_j - \Delta(\ell_0, j) \\ + \text{amenities} + \zeta_j \\ + \beta\mathbb{E}[V_{t+1}|\text{age} + 1, j]$$

Why Dynamic & Many Locations?

- Return migration and onward moves require dynamics.
- Many alternatives matter: returning is a choice among many states.



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Likelihood Function

$$\Lambda(\theta) = \sum_{n=1}^N \log\left(\sum_{\tau=1}^K \pi_{\tau} L_i(\theta_{\tau})\right) \quad (5)$$

- Observed income
- Location choice

Likelihoods

Let $\psi_{it}(\omega_i, \theta)$ correspond to likelihood of observed income of agent i in period t

$$\begin{aligned} \psi_{it}(\omega_i, \theta_\tau) \\ = \frac{1}{\sigma_\epsilon(\omega_\epsilon^i)} \phi\left(\frac{w_{it} - \mu_{l^0(i,t)} - G(X_i, a_{it}, \theta_\tau) - v(\omega_v^i(\kappa_{it}^0)) - \eta(\omega_\eta^i)}{\sigma_\epsilon(\omega_\epsilon^i)}\right) \end{aligned} \quad (6)$$

Let $\lambda_{it}(\omega^i, \theta_\tau)$ be the likelihood of the destination chosen by agent i in period t

$$\begin{aligned} \lambda_{it}(\omega^i, \theta_t) = & \rho_{h(i)}(l(i, t), \omega_v^i(\kappa_{it}^0), \omega_v^i(\kappa_{it}^1), \omega_\xi^i(\kappa_{it}^0), \\ & \omega_\xi^i(\kappa_{it}^1), a_{it}, l^0(i, t+1), \theta_\tau) \end{aligned} \quad (7)$$

Likelihoods, cont.

Likelihood of individual history for a person of type τ is

$$\begin{aligned} L_i(\theta_\tau) = & \frac{1}{n_\eta n_\epsilon n_\xi (n_v)^{N_i}} \\ & \times \sum_{\omega^i \in \Omega(N_i)} (\Pi_{t=1}^{T_i} \psi_{it}(\omega^i, \theta_\tau) \lambda_{it}(\omega^i, \theta_\tau)) \end{aligned} \quad (8)$$

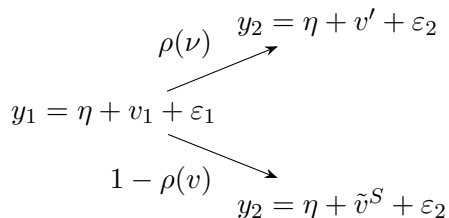
Nonparametric Identification of Choice Probabilities

- Wage matching cannot be observed directly
- We observe a match with error
- Dealt with via likelihood integrating over the distributions of the errors
- Still, heavily reliant on assumptions

Wage residual can be defined as such

$$y_{it} = \eta_i + v_{ij}(t) + \epsilon_{it} \quad (9)$$

DGP



- with probability $\rho(v)$, move and get observed wage $y_2 = \eta + v' + \varepsilon_2$
- with probability $1 - \rho(v)$, stay and get $y_2 = \eta + \tilde{v}^S + \varepsilon_2$ in which $\tilde{v}^S$ is the censored random wage derived by discarding the realized wages of movers

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Computational Preliminaries

- Use a version of Newton's algorithm with numerical derivatives
- set $\beta = 0.95$
- set $T = 40$
- set $M = 2$

What Drives Migration? (Headlines)

- **Expected income matters:** moves respond to geographic mean wages and bad current matches.
- **Costs matter:** distance penalty; strong home premium; cheaper to return.
- **Population size** of destination reduces cost (network/proximity channels).

TABLE II
INTERSTATE MIGRATION, YOUNG WHITE MEN^a

	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	$\hat{\theta}$	$\hat{\sigma}_{\theta}$
<i>Utility and cost</i>								
Disutility of moving (γ_0)	4.790	0.565	4.514	0.523	4.864	0.601	4.851	0.604
Distance (γ_1) (1000 miles)	0.265	0.182	0.280	0.178	0.311	0.187	0.270	0.184
Adjacent location (γ_2)	0.808	0.214	0.787	0.211	0.773	0.220	0.804	0.216
Home premium (α^H)	0.331	0.041	0.267	0.031	0.332	0.047	0.337	0.045
Previous location (γ_3)	2.757	0.356	2.544	0.300	3.082	0.449	2.818	0.416
Age (γ_4)	0.055	0.020	0.062	0.019	0.060	0.020	0.054	0.020
Population (γ_5) (millions)	0.653	0.179	0.653	0.178	0.635	0.177	0.652	0.179
Stayer probability	0.510	0.078	0.520	0.079	0.495	0.087	0.508	0.082
Cooling (α_1) (1000 degree-days)	0.055	0.019	0.036	0.019	0.048	0.018	0.056	0.019
Income (α_0)	0.312	0.100	—	—	—	—	0.297	0.116
Location match preference (τ_i)	—	—	—	—	0.168	0.049	0.070	0.099
<i>Wages</i>								
Wage intercept	-5.165	0.244	-5.175	0.246	-5.175	0.246	-5.168	0.244
Time trend	-0.035	0.008	-0.033	0.008	-0.033	0.008	-0.035	0.008
Age effect (linear)	7.865	0.354	7.876	0.356	7.877	0.356	7.870	0.355
Age effect (quadratic)	-2.364	0.129	-2.381	0.130	-2.381	0.130	-2.367	0.129
Ability (AFQT)	0.014	0.065	0.015	0.066	0.014	0.066	0.014	0.065
Interaction (Age, AFQT)	0.147	0.040	0.152	0.040	0.152	0.040	0.147	0.040
Transient s.d. 1	0.217	0.007	0.218	0.007	0.218	0.007	0.217	0.007
Transient s.d. 2	0.375	0.015	0.375	0.015	0.375	0.015	0.375	0.015
Transient s.d. 3	0.546	0.017	0.547	0.017	0.547	0.017	0.546	0.017
Transient s.d. 4	1.306	0.028	1.307	0.028	1.307	0.028	1.306	0.028
Fixed effect 1	0.113	0.035	0.112	0.035	0.112	0.035	0.113	0.035
Fixed effect 2	0.298	0.035	0.296	0.035	0.296	0.035	0.298	0.035
Fixed effect 3	0.936	0.017	0.934	0.017	0.934	0.017	0.936	0.017
Wage match (τ_v)	0.384	0.017	0.387	0.018	0.387	0.018	0.384	0.018
<i>Log likelihood</i>								
Exclude income: $\chi^2(1)$	-4,214.880		-4,221.426		-4,218.800		-4,214.834	
			13.09		7.93			
Exclude match preference: $\chi^2(1)$		0.09		5.25				

^aThere are 4,274 (person-year) observations, 432 individuals, and 124 moves.

Economic Magnitudes

- Moving cost equivalent $\approx$ \$300k (PV terms, order of magnitude).
- Good match + high-mean state $\Rightarrow \approx$ \$22k / year gain (1989).

Tables of Magnitudes

TABLE III
WAGE PARAMETER ESTIMATES (IN 2010 DOLLARS)

	AFQT Percentile		
	25	50	75
<i>Average wages</i>			
Age 20 in 1979	25,827	27,522	29,216
Age 20 in 1989	18,472	20,166	21,861
Age 30 in 1989	40,360	42,850	45,340
	Low	Middle	High
Location match	−8,366	0	8,366
Fixed effect support	−20,411	−6,498 −2,454	0 2,454 6,498
State means	Low (WV) 12,698	Rank 5 (OK) 14,530	Median (MO) 16,978
			Rank 45 (RI) 19,276
			High (MD) 22,229

TABLE IV
MOVING COST EXAMPLES

	γ_0	α_0	Age	Distance	Adjacent	Population	Previous	Cost
θ	4.790	0.312	0.055	0.265	0.808	0.653	2.757	
Young mover			20	1	0	1	0	\$384,743
Average mover			24.355	0.664	0.427	0.728	0.371	\$312,146

Fit & Mechanism

- Model reproduces repeat/return migration shares qualitatively.
- Negative association: higher realized income this period $\Rightarrow$ lower out-migration probability.

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What We Learn

- Migration resembles optimal search over locations.
- Both mean differences and idiosyncratic match quality drive moves.
- Structural approach enables policy counterfactuals.

Policy Angles & Extensions

- Local labor demand shocks & wage equalization dynamics.
- Subsidies or relocation assistance: who moves more?
- Household joint decisions; welfare benefit differentials; retirement migration.

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Takeaways

- Expected income plays a sizable role in interstate migration.
- High moving costs + payoff shocks reconcile low average move rates with meaningful gains.
- Model provides a tractable template for counterfactual policy analysis.

Thanks!

Questions? tate.mason@uga.edu

Roadmap

7 Appendix

Data & Sample Construction (*Appendix*)

- NLSY79 cross-section; white non-Hispanic HS grads; ages 20+; not in military.
- Attrition and consistency checks; AFQT use; state means from Census PUMS (1990).

Sample Selection

	Respondents	Person-Years
White non-Hispanic males (cross-section sample)	2,439	39,024
<i>Restrictions applied to respondents</i>		
Ever in military	-246	
High school dropouts and college graduates	-1,290	
Attended college	-130	
Older than age 20 at start of sample period	-134	
Missing AFQT score	-41	
Attend or graduate from high school at age 20	-87	
Not in labor force for more than 1 year after age 19	-44	
Location at age 20 not reported	-20	
Income information inconsistent	-1	
Died before age 30	-4	
Residence at age 14 not reported	-2	
In jail in 1993	-1	
Subtotal	-2,000	6,585
<i>Restrictions applied to periods</i>		
Delete periods after first gap in history	-1	-1,104
Delete periods before age 20	-6	-1,207
Analysis sample	432	4,274

Sample Selection — Years per person

	Respondents	Person-Years
<i>Years per person</i>		
1	14	14
2	16	32
3	19	57
4	14	56
5	14	70
6	14	84
7	13	91
8	9	72
9	34	306
10	61	610
11	53	583
12	44	528
13	45	585
14	44	616
15	38	570
Total	432	4,274

Value Function & Choice Probabilities (*Appendix*)

Bellman equation

$$V(x, \zeta) = \max_{j \in \mathcal{J}} \{ v(x, j) + \zeta_j \},$$
$$v(x, j) = u(x, j) + \beta \mathbb{E}_{x'} [\bar{V}(x') \mid x, j], \quad \bar{V}(x) = \mathbb{E}_{\zeta} [V(x, \zeta)].$$

Extreme value shocks $\Rightarrow$ logit choice probabilities

$$\rho(x, j) = \frac{\exp(v(x, j))}{\sum_{k \in \mathcal{J}} \exp(v(x, k))}, \quad \exp(\bar{V}(x)) = e^{\bar{\gamma}} \sum_k \exp(v(x, k)).$$

Limited-History State Transitions (*Appendix*)

State and transitions (with $M = 2$ recent locations)

State: $x = (\ell_0, \ell_1, \omega_{\ell_0}^{(w)}, \omega_{\ell_1}^{(w)}, \omega_{\ell_0}^{(\xi)}, \omega_{\ell_1}^{(\xi)}, a)$.

If stay: $x' = (\ell_0, \ell_1, \cdot, \cdot, \cdot, \cdot, a + 1)$.

If move to previous ℓ_1 : swap (ℓ_0, ℓ_1) and their components, $a \leftarrow a + 1$.

If move to new j : shift $(\ell_0 \leftarrow j, \ell_1 \leftarrow \ell_0)$ and draw new $(\omega_j^{(w)}, \omega_j^{(\xi)})$.

Likelihood & Identification Sketch (*Appendix*)

- Joint likelihood of income & choices; mixture over unobserved components.
- Nonparametric intuition: selection on bad current draws $\uparrow$ move probability.

Robustness (*Appendix*)

- Vary β , horizon T , include preference match, extra amenities.
- Fit diagnostics; simulated vs. observed migration rates.

Robustness (Appendix)

TABLE XIV
ESTIMATES FROM SIMULATED MIGRATION HISTORIES

	Base Model		100 Reps		<i>t</i>
	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	$\hat{\theta}$	$\hat{\sigma}_{\theta}$	
Disutility of moving	4.790	0.565	4.732	0.058	-1.00
Distance	0.265	0.182	0.272	0.015	0.42
Adjacent location	0.808	0.214	0.808	0.017	0.04
Home premium	0.331	0.041	0.333	0.004	0.42
Previous location	2.757	0.356	2.709	0.031	-1.52
Age	0.055	0.020	0.056	0.002	0.25
Population	0.653	0.179	0.648	0.017	-0.30
Stayer probability	0.510	0.078	0.517	0.008	1.00
Cooling	0.055	0.019	0.055	0.002	-0.07
Income	0.312	0.100	0.307	0.008	-0.56
Wage intercept	-5.165	0.244	-5.158	0.033	0.21
Time trend	-0.035	0.008	-0.035	0.001	0.40
Age effect (linear)	7.865	0.354	7.853	0.050	-0.23
Age effect (quadratic)	-2.364	0.129	-2.359	0.018	0.28
Ability (AFQT)	0.014	0.065	0.017	0.010	0.35
Interaction (age, AFQT)	0.147	0.040	0.142	0.007	-0.67
Transient s.d. 1	0.217	0.007	0.217	0.001	0.07
Transient s.d. 2	0.375	0.015	0.376	0.002	0.37
Transient s.d. 3	0.546	0.017	0.545	0.002	-0.84
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Fixed effect 2	0.298	0.035	0.300	0.003	0.68
Fixed effect 3	0.936	0.017	0.938	0.002	1.37
Wage match	0.384	0.017	0.383	0.002	-0.17
Log likelihood, $\chi^2(24)$	-4,214.88		-473,830.3		9.95